

HealthConnect Clinic

Project Charter

No-Show Reduction & Patient Support AI System

Prepared by: Project Manager

Sponsor: Clinic Director / Practice Manager, HealthConnect Clinic

Date: 29 August 2026

1. Executive Summary

HealthConnect Clinic is losing appointment capacity and patient engagement to missed appointments (no-shows), without knowing which patients are at risk until after they have already missed a visit. This project will build an AI-powered no-show prediction system that uses HealthConnect's appointment data — covering patient demographics, booking details, appointment history, prior no-shows, reminder activity, distance to the clinic, and waiting time — to identify at-risk patients before their appointment date. A behavioural/risk segmentation layer will sit on top of the prediction scores, enabling proactive, targeted reminder and retention action by clinic administrative and clinical staff. A Generative AI-powered patient information assistant will also be developed, scoped strictly to an approved clinic knowledge base, to reduce repetitive administrative enquiries. The solution will be deployed to a working, production-ready prototype, with an ongoing monitoring and retraining process and a formal operational handover to HealthConnect's internal IT and administrative team, so the system keeps operating and stays accurate after the project concludes.

The project runs over 20 active project weeks (approximately 5 months of active work, spanning 31 August 2026 to 19 February 2027, with December fully excluded), led by the Project Manager on behalf of the project, sponsored by HealthConnect Clinic's Director / Practice Manager, and supported by IT, Data Engineering, Data Science, Machine Learning Engineering, Generative AI, and Compliance roles. Success will be measured primarily through no-show prediction model accuracy, a measurable reduction in no-shows among flagged patients during a pilot campaign, and a successful operational handover.

2. Scope Definition

In Scope

- Data preparation and integration using the HealthConnect Appointment Dataset (representing scheduling, patient demographic, and reminder data)
- No-show prediction model development and testing
- Behavioural / risk segmentation layer built on top of prediction scores
- Dashboard / report for clinic administrative and clinical staff
- Pilot reminder / retention campaign targeting flagged at-risk patients
- Generative AI-powered patient information assistant, scoped to the approved Clinic Knowledge Base
- Production deployment of the prediction model, dashboard, and assistant
- Ongoing monitoring process for model performance and data quality
- Model retraining process and schedule
- Operational handover to HealthConnect's internal IT and administrative team

Out of Scope

- Clinical treatment planning, diagnosis, or medical decision-making
- Replacement of, or changes to, the clinic's underlying scheduling/booking software
- Clinical staff rostering or scheduling changes
- Design of a new loyalty or rewards-style patient programme
- Real-time chatbot deployment beyond the scoped information assistant
- Use of real, identifiable patient data — the project uses the fictional, anonymised dataset provided only

3. Project Manager and Sponsor Responsibilities

Project Manager Authority & Delegation

- Spending limits: may approve expenditure up to R15,000 per line item without further sign-off; amounts above this require Sponsor approval.
- Constraints on approvable changes: may approve minor scope clarifications within the agreed WBS; any change affecting the 75% model accuracy target, overall budget, or delivery date requires Sponsor sign-off through change control.
- Managing personnel performance: conducts weekly check-ins with the IT Lead, Data Engineer, Data Scientist, Machine Learning Engineer, and Generative AI Developer; performance concerns are escalated to the relevant functional line manager, with the Sponsor informed for client-side team members.
- Accessing contingency funds: may draw on the approved 10% contingency (≈R65,000) with written justification logged in the budget tracker; draws over R10,000 require Sponsor approval before release.

Sponsor Responsibilities

- Meetings attended: monthly steering committee meeting, plus ad hoc go/no-go reviews at each phase gate.
- Accessing and allocating management reserve: holds authority over any funding beyond the approved budget and contingency; requests are routed through the HealthConnect Executive Committee.
- Decisions reserved for the Sponsor: approval of scope changes, approval to proceed past each phase gate, and the decision on full production rollout after the pilot and operational handover — informed by the risk register status, budget-vs-actual reporting, pilot campaign results, and the readiness of HealthConnect's internal team to sustain monitoring and retraining.

4. Project Methodology

The project follows a predominantly predictive (Waterfall/PMBOK-aligned) methodology, reflecting its fixed scope, budget, and 20-week timeline agreed upfront. The model development work package (4.0) uses short, iterative build-test cycles within this broader plan-driven structure — a hybrid approach — since building and validating a machine learning model naturally benefits from repeated adjustment. The Generative AI assistant build (also within work package 4.0) follows the same iterative pattern. The testing, deployment, and monitoring work package (6.0) follows the same predictive structure, with its own phase-gate sign-off before production deployment.

Exceptions / Deviations from Standard Project Controls

- Project control deliverables not produced: a formal procurement plan is not required, as no external purchasing beyond existing tool/API licences is anticipated.
- Expanded project control deliverables: the risk register is more detailed than a standard template, given the project's data integration and regulatory (POPIA) compliance exposure.

5. High-Level Communication Plan

(Summary only — full detail is provided in the separate Communication Plan spreadsheet.)

- Who originates communication: the Project Manager originates internal and task-level updates; the Sponsor originates Executive-Committee-facing updates.
- What is communicated and when: weekly status updates to the Sponsor; monthly budget reports to Finance; milestone updates to all stakeholders at each phase gate.
- Where project data is sourced and stored: task tracking in the project management tool; documentation stored in the shared project folder / repository; risk, dependency, and budget logs maintained in the project's Excel registers.

6. Assumptions

- HealthConnect's appointment scheduling and reminder records share a common patient identifier, or one can be created.
- Patients have consented to data use under POPIA.
- Clinic administrative and clinical staff are available to act on model outputs during the pilot campaign.
- HealthConnect's internal IT/Data team will have the capacity to take on monitoring and retraining responsibilities after handover.

Confirmation approach: assumptions are verified in Weeks 1–2 through direct confirmation with the IT Lead (data identifiers), the Compliance Officer (consent status), and Clinic management (post-handover capacity), rather than assumed to hold throughout the project.

7. Constraints

Cost

Maximum project cost: R650,000, inclusive of a 10% contingency (≈R65,000).

Timeframe

Project must conclude by Week 20 (approximately 5 months of active work, running 31 August 2026 to 19 February 2027, with December fully excluded as a non-working period).

Scope

- Must-haves: a working no-show prediction model; at-risk patient reporting delivered to clinic administrative and clinical staff; the model, dashboard, and assistant running in production; a documented monitoring and retraining process; a completed operational handover.
- Nice-to-haves: real-time (rather than batch) score updates; segmentation criteria expanded beyond appointment history, distance, and demographics.

Regulatory Items

- Compliance with South Africa's Protection of Personal Information Act (POPIA) across all patient data handling, including in production and during ongoing monitoring.

8. Resources Required

Role	Responsibility	Time Commitment
Data Engineer	Data pipeline build and production support	≈50% dedicated time, Weeks 3–13, tapering to ≈25% through Week 19 for deployment support
Data Scientist	Model development, testing, and retraining process design	≈75% dedicated time, Weeks 3–18

Role	Responsibility	Time Commitment
Machine Learning Engineer	Reproducible pipeline architecture and version control	≈50% dedicated time, Weeks 5–19
Generative AI Developer	Patient information assistant design and build	≈50% dedicated time, Weeks 5–19
IT Lead	Systems access, technical review, production deployment, monitoring set-up	≈25% time throughout, rising to ≈50% Weeks 18–19
Compliance Officer	POPIA sign-off	As needed at phase gate reviews, including the production readiness gate
Clinic Administrative & Clinical Staff	Segment validation, dashboard training, pilot campaign execution	Light involvement, Weeks 15–17
HealthConnect Internal IT/Data Team	Onboarding to the monitoring and retraining process ahead of handover	Light involvement, Week 20

9. High-Level Risks

(Detailed further in the separate Risk Register spreadsheet.)

- Data quality issues in the appointment dataset
- Inconsistent or missing patient identifiers across records
- No-show prediction model accuracy below target
- Low staff adoption of at-risk patient flags and reminder recommendations
- Insufficient internal capacity at HealthConnect to sustain monitoring and retraining after handover
- Schedule pressure from the December office closure falling within the project timeline
- Generative AI assistant producing out-of-scope or unsafe (e.g., clinical) responses

10. Project Success Criteria

How Success Will Be Measured

- No-show prediction model accuracy scored against a held-out test dataset
- At-risk patient list delivery tracked monthly against schedule
- No-show rate among the pilot's flagged segment measured against a control group
- Production uptime and monitoring/retraining process adherence post-deployment

Definition of Acceptable Success

- No-show prediction model achieves at least 75% prediction accuracy on test data
- At-risk patient list is delivered to clinic staff on a monthly cadence with no missed cycles
- Pilot reminder campaign reduces no-shows in the flagged segment by at least 10% versus the control group
- Model, dashboard, and assistant are deployed to production and formally handed over, with HealthConnect's internal team able to run the monitoring and retraining process unassisted

11. Approval

This charter authorises the Project Manager to proceed with project initiation as described above.

Sponsor Signature: _____ Date: _____

Project Manager Signature: _____ Date: _____